

MSTAR PRACTICE WORKSHEET**Lesson 3-9: Equivalent Ratios**

Grade 6 Mathematics · Standard 6.AT.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: C**

Which ratio is equivalent to 2:5?

- ☐ A 2:10
- ☐ B 3:6
- ☒ C 4:10
- ☐ D 5:2

Multiply both parts by 2: $2 \times 2 = 4$ and $5 \times 2 = 10$, so 4:10 is equivalent to 2:5.

Question 2 **SELECTED RESPONSE — CORRECT: A**

A recipe uses 4 eggs for every 6 cups of flour. Which ratio is equivalent?

- ☒ A 2:3
- ☐ B 4:3
- ☐ C 6:4
- ☐ D 8:10

Divide both by 2: $4 \div 2 = 2$ and $6 \div 2 = 3$. The ratio 2:3 is equivalent to 4:6.

Question 3 SELECTED RESPONSE — CORRECT: B

Which ratio is NOT equivalent to 6:9?

- ☐ A 2:3
- ☒ B 3:5
- ☐ C 12:18
- ☐ D 18:27

6:9 simplifies to 2:3. Check each: $2:3 \checkmark$, $12:18 = 2:3 \checkmark$, $18:27 = 2:3 \checkmark$. But 3:5 does not simplify to 2:3.

Question 4 SELECTED RESPONSE — CORRECT: B

What is the simplest form of the ratio 15:25?

- ☐ A 1:5
- ☒ B 3:5
- ☐ C 5:3
- ☐ D 15:25

The GCF of 15 and 25 is 5. Divide both by 5: $15 \div 5 = 3$ and $25 \div 5 = 5$. The simplest form is 3:5.

Question 5 SELECTED RESPONSE — CORRECT: C

Marcus earns \$45 for every 3 hours of work. How much does he earn in 8 hours?

- ☐ A \$90
- ☐ B \$105
- ☒ C \$120
- ☐ D \$135

Unit rate: $\$45 \div 3 = \$15/\text{hour}$. In 8 hours: $\$15 \times 8 = \120 .

Question 6 **SELECTED RESPONSE — CORRECT: D**

Chef Montoya's dressing uses 3 spoons of oil to 4 spoons of vinegar (3:4). Which recipe card shows an EQUIVALENT ratio?

- ☐ A 3:8
- ☐ B 4:3
- ☐ C 6:7
- ☒ D 9:12

Multiply both parts of 3:4 by 3: $3 \times 3 = 9$ and $4 \times 3 = 12$, so 9:12 is equivalent to 3:4.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 **TWO-PART QUESTION****Part A — correct answer: B**

Solve the proportion $4/10 = x/25$ for x .

- ☐ A 8
- ☒ B 10
- ☐ C 19
- ☐ D 62.5

The scale factor from 10 to 25 is 2.5, so $x = 4 \times 2.5 = 10$. Cross products confirm it: $4 \times 25 = 100$ and $10 \times 10 = 100$.

- **A:** That doubles the 4. The denominator went from 10 to 25, a factor of 2.5 rather than 2.
- **C:** That adds the 15 that was added to the denominator. Proportions scale by multiplying.
- **D:** That scales the 25 instead of the 4. Apply the factor 2.5 to the numerator.

Part B — correct answer: B

How do cross products show that two ratios are equivalent?

- ☐ A If $a + d$ equals $b + c$, the ratios are equivalent.
- ☒ B If $a/b = c/d$, then $a \times d$ equals $b \times c$.
- ☐ C Cross products work only when all four numbers are whole.
- ☐ D Cross products compare the two numerators only.

Multiplying both sides of $a/b = c/d$ by $b \cdot d$ gives $a \cdot d = b \cdot c$, so equal cross products mean the ratios are equivalent.

Question 8

SELECT ALL THAT APPLY — CORRECT: A, B, D

Select ALL pairs of ratios that are equivalent:

☒ A $2/3$ and $8/12$

☒ B $5/8$ and $10/16$

☐ C $3/4$ and $4/5$

☒ D $6/9$ and $2/3$

☐ E $7/10$ and $14/21$

A ($2 \times 12 = 24 = 3 \times 8$), B ($5 \times 16 = 80 = 8 \times 10$), and D ($6 \times 3 = 18 = 9 \times 2$) are equivalent. C fails: $3 \times 5 = 15$ but $4 \times 4 = 16$. E fails: $7 \times 21 = 147$ but $10 \times 14 = 140$.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A store sells 3 pens for \$2. Another store sells 9 pens for \$6. Are these equivalent ratios? If so, prove it using at least two different methods (simplifying, scaling, or cross-multiplying).

Model answer: Yes, they are equivalent. Method 1 (simplify): $9:6 \div 3 = 3:2$, which matches $3:2$. Method 2 (cross-multiply): $3 \times 6 = 18$ and $2 \times 9 = 18$. Since both cross products are equal, the ratios are equivalent.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.